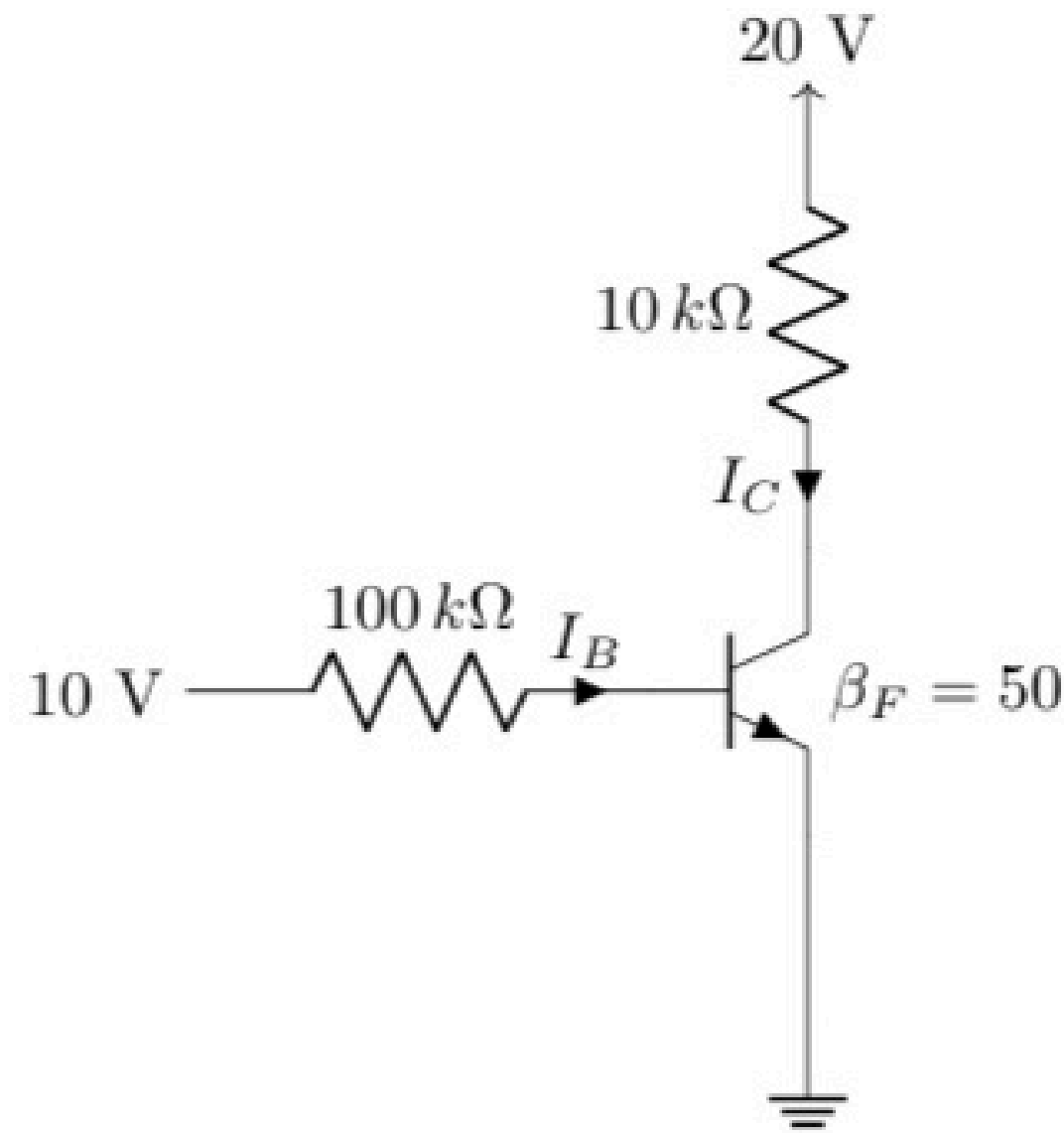


Problem 1:



Assuming the given BJT is in forward active mode—

(a)	Find I_B , I_C & I_E .
(b)	Find V_C .
(c)	Verify if the BJT is in forward active mode.

Solution:

(a) In forward active mode, $V_{BE} = 0.7\text{ V}$. Since $V_E = 0\text{ V}$, $V_B = 0.7\text{ V}$. So,

$$I_B = \frac{10 - 0.7}{100k} = 0.093\text{ mA}$$

$$\therefore I_C = \beta_F I_B = 50 \times 0.093 = 4.65\text{ mA}$$

$$\therefore I_E = I_C + I_B = 4.743\text{ mA}$$

(b) $V_C = 20 - I_C \times 10 = -26.5\text{ V}$

(c) $V_{CE} = -26.5 - 0 = -26.5 < 0.2\text{ V}$. So, the BJT is not operating in forward active mode and our assumption was wrong.

Problem 2:

Assuming saturation mode for the BJT in **Problem 1**—

(a)	Find I_B , I_C & I_E .
(b)	Verify if the BJT is in saturation mode.

Solution:

(a) For saturation mode, $V_{BE} = 0.8\text{ V}$. Since $V_E = 0\text{ V}$, $V_B = 0.8\text{ V}$. So,

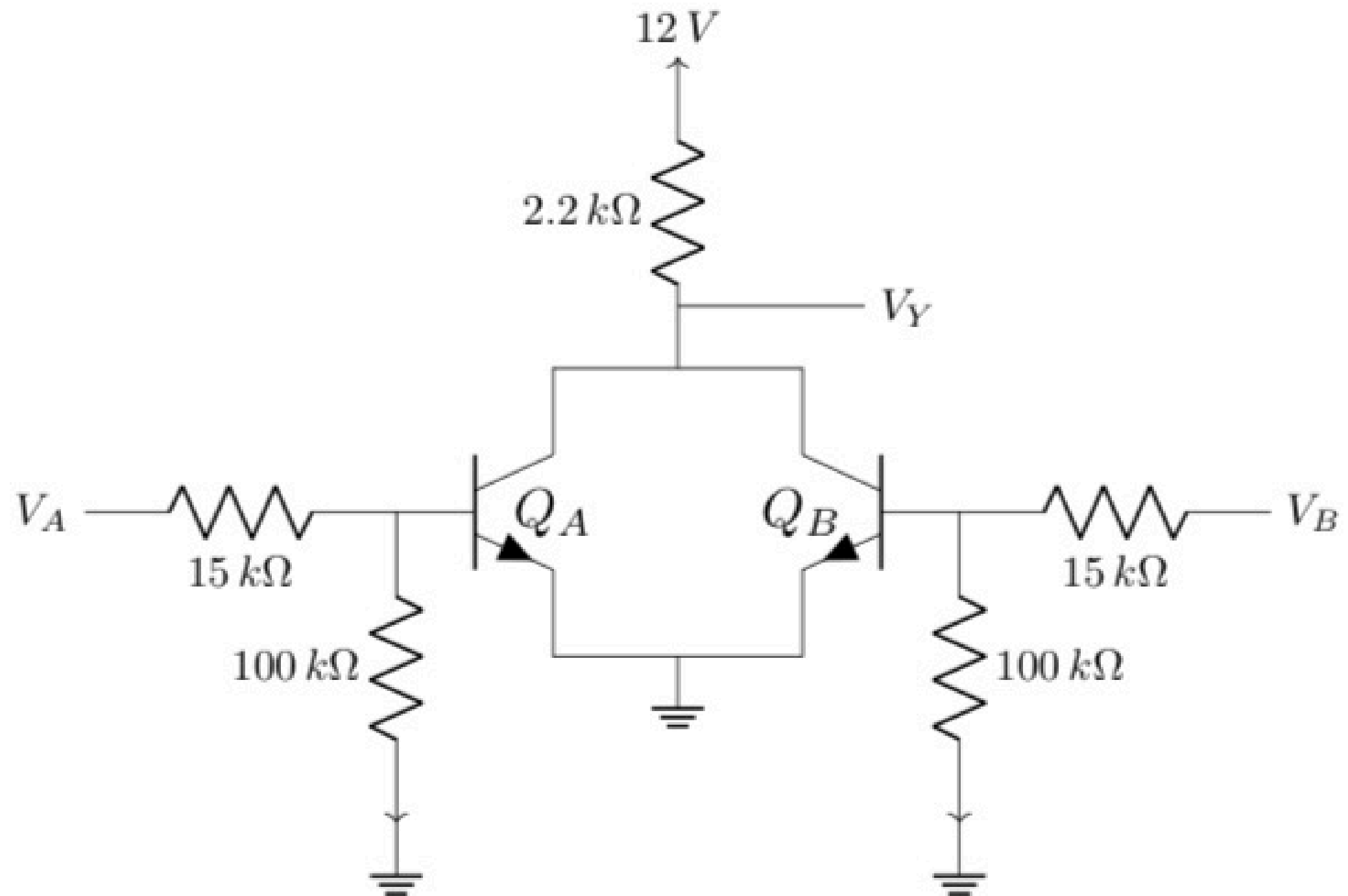
$$I_B = \frac{10 - 0.8}{100} = 0.092\text{ mA}$$

Also, $V_{CE} = 0.2\text{ V}$. So, $V_C = 0.2\text{ V}$.

$$\therefore I_C = \frac{20 - 0.2}{10} = 1.98\text{ mA}$$

$$\therefore I_E = I_C + I_B = 2.072\text{ mA}$$

(b) $\beta_{forced} = \frac{I_C}{I_B} = 21.52 < \beta_F$

Problem 3:

For both Q_A & Q_B , $\beta_F = 30$.

(a)	Determine the output voltage for all input cases.
(b)	What are the high & low thresholds of the output voltage?
(c)	If the '100 kΩ' of ' Q_B ' is doubled, find the new I_B of Q_B . Will it still satisfy saturation mode conditions for Q_B in case of (1,1) & (0,1)?

Solution:

(a) Case (0,0): Q_A & Q_B both in cut-off. So, $I_{RC} = 0$. Thus, $V_Y = 12\text{ V}$.

Case (0,1): Q_A in cut-off & Q_B in saturation. So, $V_Y = V_C = 0.2 V$.

Case (1,0): Similar as case (0,1), $V_Y = 0.2 V$.

Case (1,1): Q_A & Q_B are both in saturation. So, $V_Y = V_C = 0.2 V$.

(b) $V_{OH} = 12 V$, $V_{OL} = 0.2 V$.

$$(c) I_{B_{new}} = \frac{12 - 0.8}{15} - \frac{0.8}{200} = 0.743 mA$$

$$\text{For (0,1): } I_{C_B} = I_{R_C} = \frac{12 - 0.2}{2.2} = 5.363 mA$$

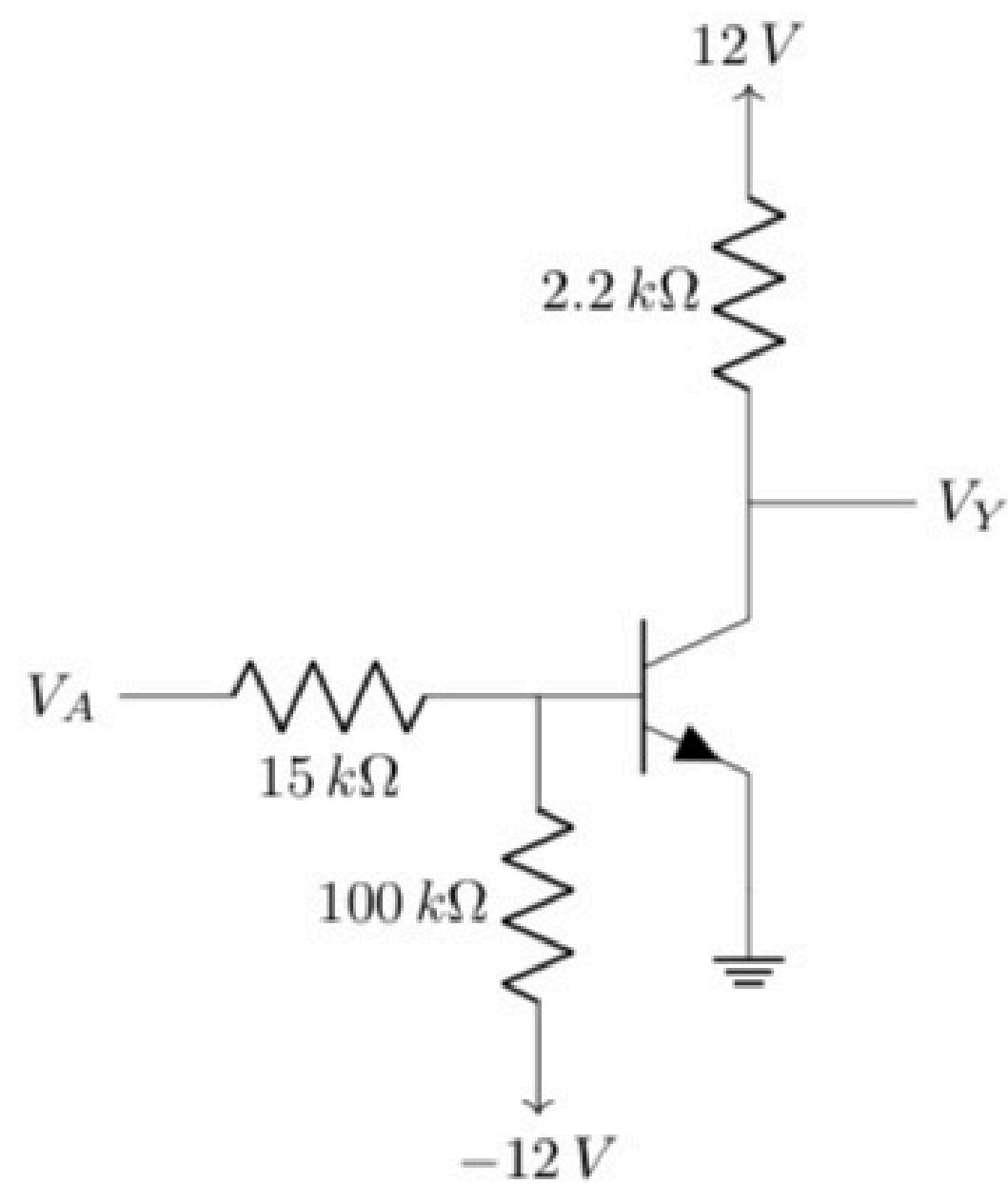
$$\beta_{forced} = \frac{5.363}{0.743} = 7.22 < \beta_F$$

$$\text{For (1,1): } I_{C_B} = \frac{I_{R_C}}{2} = 2.682 mA$$

$$\beta_{forced} = \frac{2.682}{0.743} = 3.6 < \beta_F$$

So, saturation mode conditions are satisfied.

Problem 4:



Find the maximum & average power dissipation of the RTL NOT gate shown. [Assume low input = 0.2 V, high input = 12 V]

Solution:

$$\text{Case (0): } P_0 = (0.2 - (-12)) \times \frac{0.2 - (-12)}{15 + 100} = 1.29 \text{ mW}$$

$$\text{Case (1): } P_1 = (12 - 0.8) \times \frac{12 - 0.8}{15} + (0.8 - (-12)) \times \frac{0.8 + 12}{100} + (0.8 - 0) \times \left(\frac{12 - 0.8}{15} - \frac{0.8 - (-12)}{100} \right) +$$

$$(12 - 0) \times \frac{12 - 0.2}{2.2} = 74.86 \text{ mW}$$

$$\therefore P_{max} = 74.86 \text{ mW}$$

$$\therefore P_{avg} = \frac{74.86 + 1.29}{2} = 38.075 \text{ mw}$$

Problem 5:

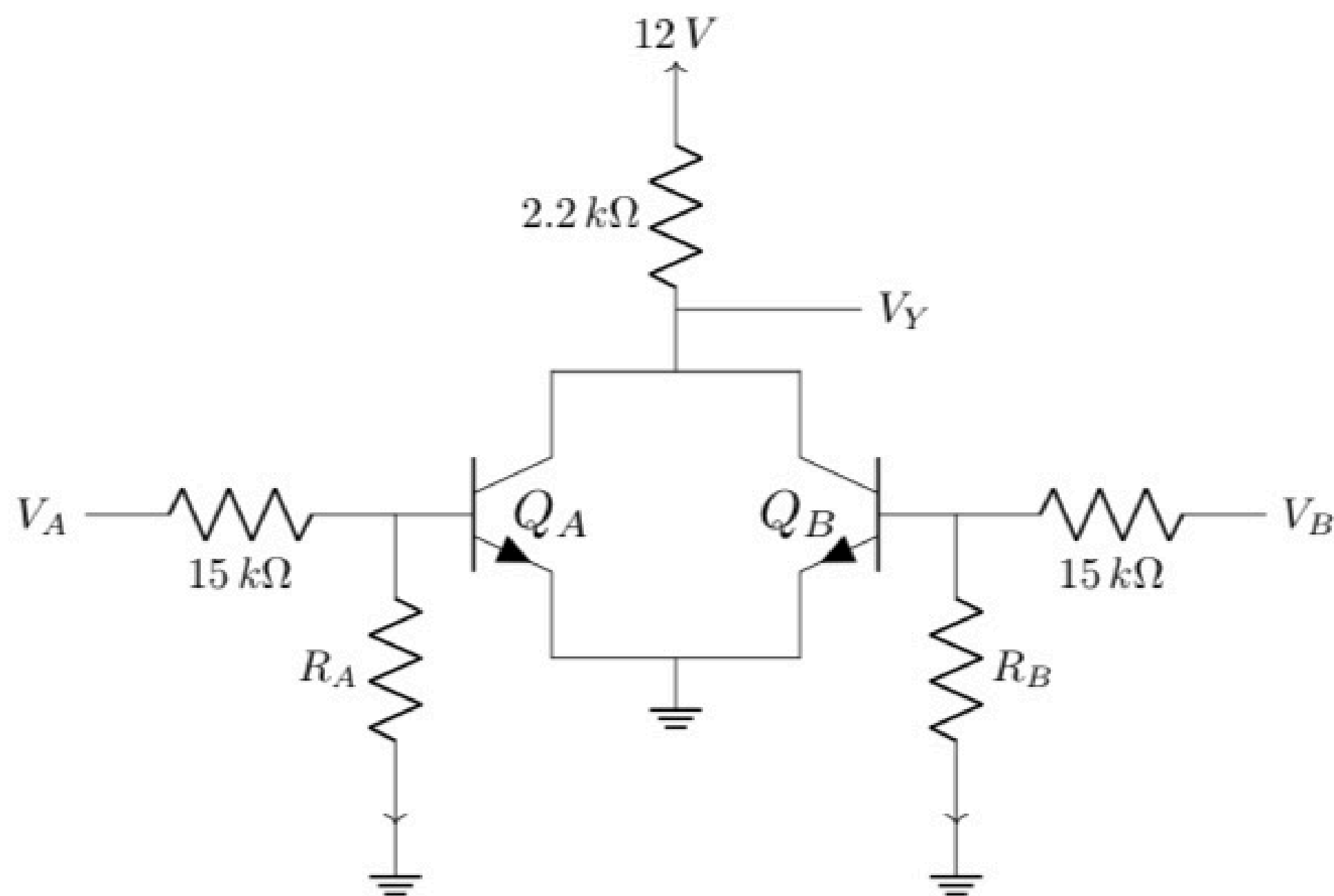
For the RTL inverter in **Problem 4**, if the '-12 V' was replaced with a ground, find the ratio of maximum and minimum dissipated power.

Solution:

$$\text{Case (0): } P_0 = (0.2 - 0) \times \frac{0.2 - 0}{15 + 100} = 0.35 \mu\text{W}$$

$$\text{Case (1): } P_1 = (12 - 0) \times \frac{12 - 0.8}{+} (12 - 0) \times \frac{12 - 0.2}{2.2} = 73.32 \text{ mW}$$

$$\therefore \frac{P_{max}}{P_{min}} = \frac{P_1}{P_0} = 209496.1 : 1$$

Problem 6:

For the given circuit, if $R_A = R_B$ —

(a)	If the power dissipation for the logic case (0,0) is 0.5 mW , find the value of the resistances.
(b)	Find a new value of R_C to halve the value of maximum power dissipation of the circuit.

Solution:

(a) Case (0,0):

$$P_{(0,0)} = 2 \times (0.2 - 0) \times \frac{0.2 - 0}{15 + R_A} = 0.25 \times 10^{-6} \Rightarrow R_A = 145 \text{ k}\Omega$$

$$(b) P_{max} = P_{(1,1)} = 2 \times (12 - 0) \times \frac{12 - 0.8}{15} + (12 - 0) \times \frac{12 - 0.2}{2.2} = 82.28 \text{ mW}$$

$$P'_{max} = 2 \times 82.28 = 164.56 \text{ mW}$$

$$\text{Now, } P'_{max} = 2 \times (12 - 0) \times \frac{12 - 0.8}{15} + (12 - 0) \times \frac{12 - 0.2}{R'_C} \Rightarrow R'_C = 0.966 \text{ k}\Omega$$